# Lecture 5: The Geometry of Linear Optimization

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# Announcements

- Homework 1 due next Wednesday (2/15)

# Recap

In the first two weeks, we have learned how to

- Formulate linear optimization problems
- Transform an LP into a standard form
- Use software to solve linear optimization

Next..

- How to solve linear optimization problem
- We will start with some basic properties of LP

# Starting Point: Graphical Solutions to LP

It is very helpful to study a small LP from a graphical point of view.

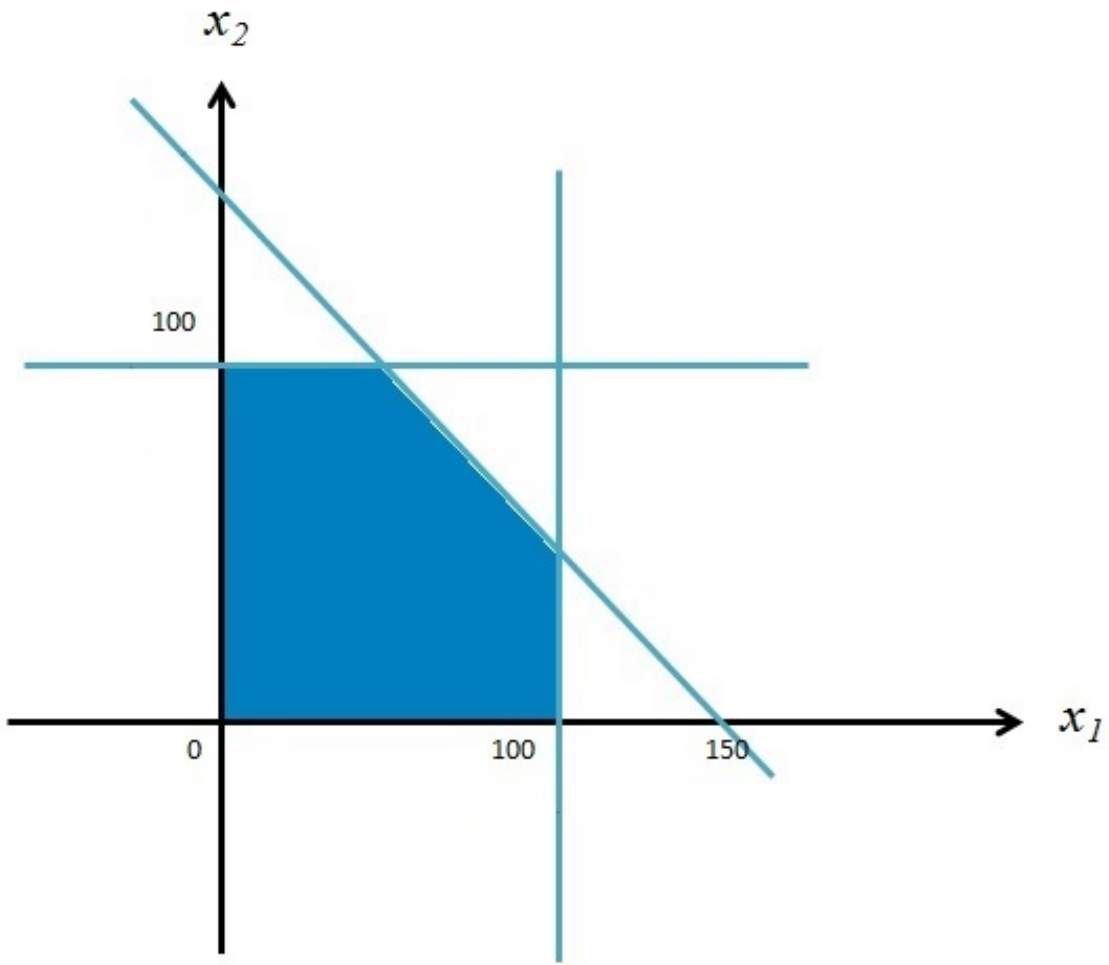
Recall the production problem:

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

How can we solve this from a graph?

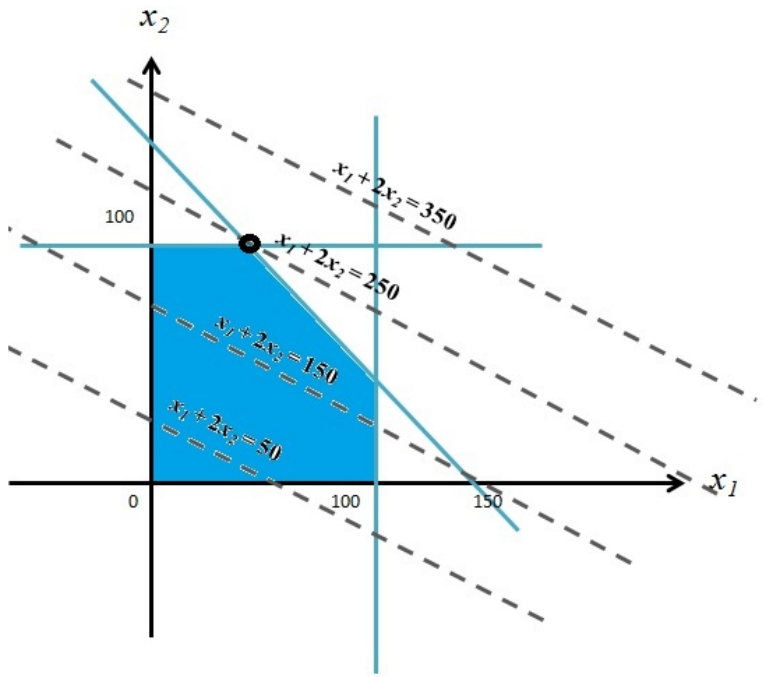
# Solve LP from Graph

We first draw the feasible region.



# To Maximize $x_1 + 2x_2$...

Then we draw the function $x_1 + 2x_2 = c$ for different values of $c$.




- The optimal solution is the highest one among these lines that touch the feasible region
- The coordinates: $(50, 100)$. Objective value: 250

# Some Observations

- The feasible region of LP is a polygon 多边形
- The optimal solution tends to be at a corner of the feasible region
- Some constraints are *active* at the optimal solution ($x_2 \leq 100$, $x_1 + x_2 \leq 150$), some are not ($x_1 < 100$).

Next we will formalize these observations

# Some Definitions: Polyhedron

## Definition

A *polyhedron* is a set that can be described in the form

$$\{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} \geq \mathbf{b}\}$$

where $A$ is an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$.

- Recall that in the standard form of LP, the feasible set is

$$A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0$$

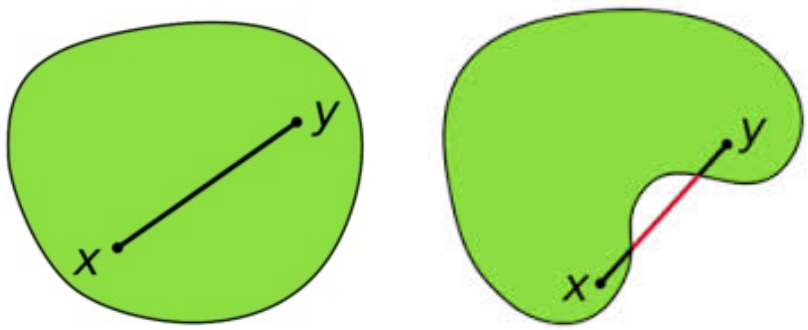
Is this a polyhedron? Why?

- Yes, we can write it as $A\mathbf{x} \geq \mathbf{b}$, $A\mathbf{x} \leq \mathbf{b}$, $\mathbf{I}\mathbf{x} \geq 0$ where $\mathbf{I}$ is the identity matrix.

# Convex Sets and Convex Combinations

## Definition

A set $S \subseteq \mathbb{R}^n$ is *convex* if for any $\mathbf{x}$, $\mathbf{y} \in S$, and any $\lambda \in [0, 1]$, $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S$.




(a) Convex set (b) Non-convex set

## Definition

For any $\mathbf{x}_1, ..., \mathbf{x}_n$ and $\lambda_1, ..., \lambda_n \geq 0$ satisfying $\lambda_1 + \cdots + \lambda_n = 1$, we call $\sum_{i=1}^{n} \lambda_i \mathbf{x}_i$ a convex combination of $\mathbf{x}_1, ..., \mathbf{x}_n$.

# Extreme Points

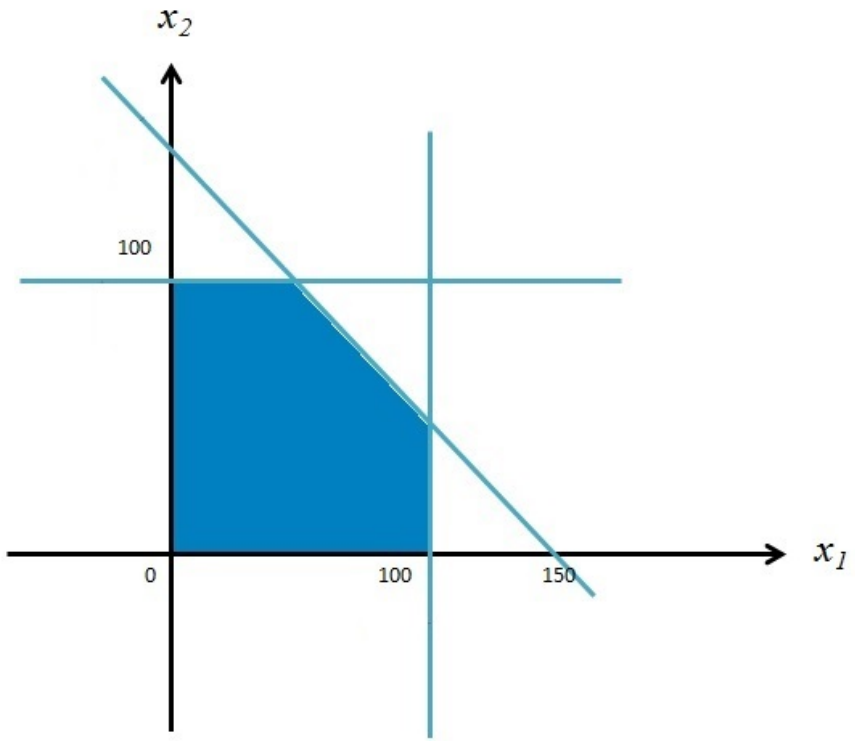
Remember we noticed that in an LP, the optimal solution tends to be on one of the corners of the feasible region. We first formalize this notion:

## Definition

Let $P$ be a polyhedron. A point $\mathbf{x} \in P$ is said to be an extreme point of $P$ if we cannot find two vectors $\mathbf{y}, \mathbf{z} \in P$, both different from $\mathbf{x}$, and a scalar $\lambda \in [0, 1]$, such that $\mathbf{x} = \lambda \mathbf{y} + (1 - \lambda)\mathbf{z}$

- That is, $\mathbf{x}$ cannot be represented as a convex combination of other points in $P$.
- We sometimes call the extreme point the *vertex*, or *corner* of the polyhedron

# Example




How many extreme points are there in this feasible region?

- Answer: 5

# Extreme Points

We just gave the definition of the extreme points/vertices. However, this doesn't tell us how to find it. We want to have a good way of finding those points.

In the following

- We will introduce an algebraic way to represent the extreme points
- We will show that it is sufficient to look among extreme points to solve a linear optimization problem
- Finally it will lead to an algorithm (the simplex algorithm) for solving LPs

# Find Extreme Points of an LP

In the following, we consider LP in its standard form:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

$m > n$ usually no solution
$m = n$ unique solution
no need to maximize

where $\mathbf{x} \in \mathbb{R}^n$, $A$ is an $m \times n$ matrix $(m < n)$ and $\mathbf{b} \in \mathbb{R}^m$.

## Assumption

*$A$ has linearly independent rows (or equivalently $A$ has full rank $m$)*

What if it is not satisfied?
- consistent: remove the redundant ones
- inconsistent: no feasible solution

- Then it means that either there is a redundant constraint (in which case one can remove it), or the constraints are not self-consistent (in which case there is no feasible solution)

# Find Extreme Points of an LP

Now we study the extreme points of an LP in its algebraic form.

**Definition**

We call $\mathbf{x}$ a *basic solution* of the LP if and only if

1. $A\mathbf{x} = \mathbf{b}$
2. There exist indices $B(1), ..., B(m)$ such that the columns $A_{B(1)}, ..., A_{B(m)}$ are linearly independent, and $x_i = 0$ for $i \neq B(1), ..., B(m)$

index 复数

# Finding a Basic Solution

# independent cols = # independent rows

Procedures to find a basic solution:

1. Choose any $m$ independent columns of $A$: $A_{B(1)}, ..., A_{B(m)}$
2. Let $x_i = 0$ for all $i \neq B(1), ..., B(m)$
3. Solve the equation $A\mathbf{x} = \mathbf{b}$ for the remaining $x_{B(1)}, ..., x_{B(m)}$.

Since $A_{B(1)}, ..., A_{B(m)}$ are linearly independent, the last step must produce a unique solution.

## Remark

*Basic solution of an LP only depends on its constraints, it has nothing to do with the objective function.*

# Basic Solutions of LP

We write

$$A_B = \begin{bmatrix} | & | & & | \\ A_{B(1)} & A_{B(2)} & \cdots & A_{B(m)} \\ | & | & & | \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix}$$

Remember that $A_{B(i)}$'s are linearly independent, therefore $A_B$ is invertible and $\mathbf{x}_B = A_B^{-1}\mathbf{b}$

We call $B = \{B(1), ..., B(m)\}$ the basic indices for this basic solution, $A_{B(1)}, ..., A_{B(m)}$ the basic columns, $A_B$ the basis matrix and $x_{B(1)}, ..., x_{B(m)}$ the basic variables

- We call the remaining indices the non-basic indices, the remaining columns of $A$ the non-basic columns and the remaining variables the non-basic variables.

# Quiz

How many non-zeros could one have in a basic solution (assuming there are $m$ constraints)?

- No more than $m$
- Could be anything between 0 to $m$, but typically it is $m$

How many basic solutions can one have for a linear program with $m$ constraints and $n$ variables?

- At most $C(n,m) = \frac{n!}{m!(n-m)!}$ (Combination number)
  ($n$个里选$m$个)
- Therefore for a finite number of linear constraints, there can only be a finite number of basic solutions

# Basic Feasible Solutions

**Definition**

If a basic solution $\mathbf{x}$ also satisfies that $\mathbf{x} \geq 0$, then we call it a *basic feasible solution* (BFS).

To find a BFS

- First find a basic solution $\mathbf{x}$
- Check if $\mathbf{x} \geq 0$

**Theorem**

*For the standard LP polyhedron* $\{\mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$, *the followings are equivalent:*

1. $\mathbf{x}$ *is an extreme point*
2. $\mathbf{x}$ *is a basic feasible solution*

# Example

Recall the production problem:

$$
\begin{array}{rrrrl}
\text{maximize} & x_1 & +2x_2 & & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The standard form:

$$
\begin{array}{rrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & \\
\text{subject to} & x_1 & & +s_1 & & & = 100 \\
& & 2x_2 & & +s_2 & & = 200 \\
& x_1 & +x_2 & & & +s_3 & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & \geq 0
\end{array}
$$

# Example Continued

We can write the feasible set by $\{\mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$. where

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix}$$

Choose three independent columns of $A$, e.g., the first three (i.e., $B = \{1, 2, 3\}$), we get the corresponding basic solution is

$$\mathbf{x}_B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \\ 50 \end{bmatrix}$$

That is $x_1 = 50, x_2 = 100, s_1 = 50$. Therefore $(50, 100, 50, 0, 0)$ is a basic feasible solution. One can find other basic feasible solutions by choosing other sets of columns.

# Example Continued

We can list the basic (feasible) solutions

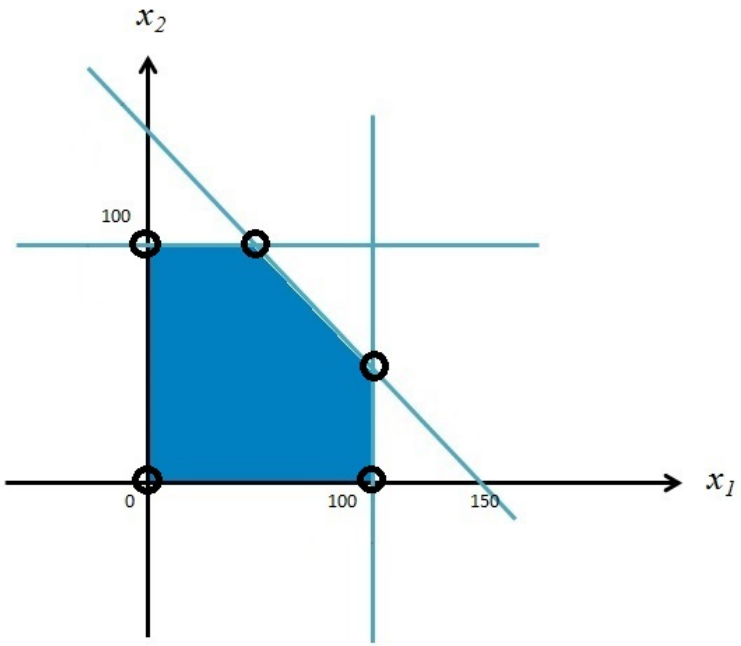
| Indices | $\{1,2,3\}$ | $\{1,2,4\}$ | $\{1,2,5\}$ | $\{1,3,4\}$ |
|---|---|---|---|---|
| Solution | (50, 100, 50, 0, 0) | (100, 50, 0, 100, 0) | (100, 100, 0, 0, -50) | (150, 0, -50, 200, 0) |
| Status | BFS | BFS | Basic but not feasible | Basic but not feasible |

| Indices | $\{1,4,5\}$ | $\{2,3,4\}$ | $\{2,3,5\}$ | $\{3,4,5\}$ |
|---|---|---|---|---|
| Solution | (100, 0, 0, 200, 50) | (0, 150, 100, -100, 0) | (0, 100, 100, 0, 50) | (0, 0, 100, 200, 150) |
| Status | BFS | Basic but not feasible | BFS | BFS |

The other two choices $\{1,3,5\}$ and $\{2,4,5\}$ lead to dependent basic columns (therefore no basic solutions can be obtained)

# Verify..

They indeed correspond to all the corners of the feasible sets.



# Proof Sketch

We show how to prove $1 \Rightarrow 2$ (the other part can be found in Chapter 2 of Bertsimas's book). First, if $\mathbf{x}$ is an extreme point, then it must be a feasible solution. In the following, we prove by contradiction that $\mathbf{x}$ must be a basic solution as well.

Let $B = \{B(1), ..., B(k)\}$ be the set of indices such that $x_i > 0$. If $\mathbf{x}$ is not a basic solution, then it must be that (why?)

- Columns $A_{B(1)}, ..., A_{B(k)}$ are linearly dependent.

Say $\alpha_1 A_{B(1)} + \cdots + \alpha_k A_{B(k)} = 0$, where $(\alpha_1, ..., \alpha_k) \neq 0$.

Now since $x_i > 0$, $\forall i \in B$, we can choose a small $\epsilon$ such that both

$$\bar{\mathbf{x}} = \begin{cases} x_i + \epsilon\alpha_j & i = B(j) \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \tilde{\mathbf{x}} = \begin{cases} x_i - \epsilon\alpha_j & i = B(j) \\ 0 & \text{otherwise} \end{cases}$$

are still feasible solutions.

Then we have $\mathbf{x} = 0.5\bar{\mathbf{x}} + 0.5\tilde{\mathbf{x}}$, which contradicts with that $\mathbf{x}$ is an extreme point.



# Basic Feasible Solutions

**Theorem**

*(LP fundamental theorem) Given a linear optimization problem where A has full row rank m*

1. *If there is a feasible solution, there is a basic feasible solution;*
2. *If there is an optimal solution, there is an optimal solution that is a basic feasible solution.*

**Corollary**

*In order to find an optimal solution, we only need to look among basic feasible solutions.*

**Corollary**

*If an LP with m constraints (in the standard form) has an optimal solution, then there must be an optimal solution such that there is no more than m positive entries.*

# Proof Sketch

We first prove part (i). Assume there is a feasible solution $\mathbf{x}$. Let $B$ denote the index set for which $\mathbf{x}$ is positive, and let $k$ be the size of $B$. Then we show that if $\mathbf{x}$ is not a basic solution, we can always find a solution that has an index set with size less than $k$. By repeatedly doing this we must be able to find a basic feasible solution.

To show that, if $\mathbf{x}$ is not a basic solution, then it must be that $A_i$, $i \in B$ are linearly dependent. Suppose $\sum_{i \in B} \alpha_i A_i = 0$ where at least one $\alpha_i$ is positive (we can always achieve that). And we define a vector $\boldsymbol{\alpha}$ by

$$\boldsymbol{\alpha} = \begin{cases} \alpha_i & i \in B \\ 0 & \text{otherwise} \end{cases}.$$

Then we consider $\mathbf{x} - \epsilon\boldsymbol{\alpha}$. There must exist an $\epsilon$ such that it is still feasible and that there is one more 0 entry in $\mathbf{x} - \epsilon\boldsymbol{\alpha}$. Thus part (i) is proved.

# Proof Sketch Continued

Now we consider part (ii). We continue to use the arguments for part (i). Let $\mathbf{x}$ be an optimal solution with $k$ positive entries. If $\mathbf{x}$ is not a basic solution, then we show that we can always find an optimal solution with fewer than $k$ positive entries (or reach a contradiction). By repeatedly doing this, we must be able to find an optimal basic feasible solution.

We continue to define the same $\boldsymbol{\alpha}$. We claim that it must be that $\mathbf{c}^T\boldsymbol{\alpha} = 0$, otherwise, one can construct a better solution than $\mathbf{x}$ by considering $\mathbf{x} + \epsilon\boldsymbol{\alpha}$ and $\mathbf{x} - \epsilon\boldsymbol{\alpha}$ for some small $\alpha$, which contradicts with the optimality of $\mathbf{x}$.

If $\mathbf{c}^T\boldsymbol{\alpha} = 0$, then $\mathbf{x} - \epsilon\boldsymbol{\alpha}$ are still optimal solutions. And by previous arguments, we can find one optimal solution with fewer positive entries. □

# Basic Feasible Solutions

Now we know that we only need to search among basic feasible solutions for the optimal solution.

How to search among the basic feasible solutions?

- One may suggest to list all the basic feasible solutions and compare their objective values. However, there are too many of them.
- For a linear optimization with $m$ constraints and $n$ variables, how many basic feasible solutions it may have?
- $C(n, m)$.. If $n = 1000$, $m = 100$, then the value is $10^{143}$..

# Simplex Method

Therefore we need a smarter way to find the optimal solution.

- Simplex method

The simplex method proceeds from one BFS (a corner point of the feasible region) to a neighboring one, in such a way as to continuously improve the value of the objective function until reaching optimality.

- We need to define what it means by *adjacent* or *neighboring* solution
- We need to design an efficient way to find (and move to) the neighboring BFS (e.g., we should try to avoid taking matrix inversions every time)
- We need to design a valid stopping criterion

We will discuss these in the next lecture